

SAFETY MEMORY EQUATION

M02023_next = (M02023_old OR fresh M02002) AND raw_normal AND NOT first_scan
Any raw abnormal input clears permission; restoring the input alone does not rearm.

MAIN START PERMISSION	
Condition	Before START
Safety / door	M01000=1 / M01007=1
Environment	M0100C=1 from receiver
Mode / state	D00053=0 / D00100=10
Cycle choice /STOP	D01220 is0/1; M00046=0
Alarm / ATE DONE	D00310=0 / D00021=0
Position	Clamp open, no test/outfeed product
Gates	All three gate feedback bits=0

RECOVERY RESPONSIBILITY

M02002: virtual safety rearm only.
M02003: clear interrupted plant model when outputs OFF.
M00202: process RESET only after recovery conditions.
M00200: separate new START after recovery.
Neither popup closing nor environment recovery starts the machine.

This is functional safety-relay simulation. Personnel protection must be implemented by an independently assessed external safety circuit.

NO	DATE	REVISION
R0	2026-10-22	FUNCTION RECORD

DWG. TITLE

SEMICON SORTER PROJECT
Final Test and Multi-Bin Sorting Cell

VIRTUAL SAFETY AND STARTUP CONDITIONS

DRAWN BY	DATE
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005

SHEET NO
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